Gujarat Police Hackathon

2-Minute Project Presentation Script

Today, we are building an **AI-powered intelligent surveillance system for Gujarat Police**.

The problem is simple: there can be thousands of video streams from different locations and vendors, but it is impossible for police personnel to continuously monitor every stream.

Our solution is to build a centralized **Video Management System, or VMS**, that can manage this large video network and turn it into an intelligent, automated surveillance system.

First, we create a centralized **vendor and video-source management system**. Video sources from different vendors can be registered in our database along with their location, district, state, stream URL, protocol and operational status.

The video streams are then managed through our **Video Management System**, with a high-performance networking layer being developed using **C++** for efficient communication and scalability.

On top of the video streams, we introduce our **AI engine**.

The AI can detect and track vehicles in real time and count how many vehicles are passing through a particular road. Because we use tracking, the same vehicle is not counted repeatedly across different frames.

But vehicle counting is only the beginning.

Our future AI modules can detect **accidents, abnormal activity, wrong-way driving, selected traffic-rule violations and other important events**.

Whenever an important event is detected, it becomes a structured event containing information such as the video source, location, timestamp, event type and evidence.

And this is where our system becomes more useful for police operations.

We plan to build a dedicated **Police Mobile Application**.

Instead of requiring officers to continuously watch CCTV screens, important emergency events can generate alerts directly on their phones.

For example: **“Possible accident detected — Video Source 1024 — XYZ Road — High Priority.”**

The officer can open the alert, view the location and available evidence, access the relevant video stream, acknowledge the incident and update its response status.

So our complete workflow becomes:
Video Management System → **AI** → **Detection** → **Alert** → **Police Officer** → **Response.**

In the future, we can expand this system with **license-plate recognition, cross-video-source vehicle tracking, intelligent search, geographic dashboards, distributed AI processing and support for tens of thousands of video streams**.

Our goal is not just to build a CCTV monitoring system.

Our goal is to build a **scalable video-to-intelligence platform that helps police detect important situations faster and respond more effectively.**